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Expectations of Software Product Managers: An Empirical Evaluation of Software Professionals' Perspectives

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The role of the software product manager is often ambiguous and potentially contentious, with tensions often arising between the software product manager and the software team. The goal of this work is to uncover software product manager role expectations from three perspectives: their own, that of the software team, and that of the software companies. We deployed a survey targeting software professionals (n=80, 22 companies) to uncover software product manager role expectations and the typical questions they answer, providing direct and indirect lenses into software product manager activities. Through qualitative analysis, we discovered that software product managers must be PRODUCT EXPERTS, USER ADVOCATES, and COMMUNICATION GURUS. From the companies' perspective, we analyzed 37 job descriptions to generalize our findings. We also identify that communication and competency issues lead to frequent team frustration. Our findings provide clarity on the software product manager role directly through employees' and employers' perspectives, indirectly from the questions they should answer, and through eight friction points when working with a software team. We synthesize these results across the software development lifecycle in the SENSE framework, providing a comprehensive view of software product manager role expectations and points of friction with software teams.

CCS Concepts: • **Software and its engineering** → **Software creation and management**; **Collaboration in software development**.

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1 Introduction

Modern software teams require the coordination of many individuals, including software engineers, engineering managers, data scientists, user experience researchers, designers, and, the focus of this research, software product managers. With a shift to a more customer-centric and iterative delivery model [26], embedding product managers in software teams is becoming essential [10, 12], and the job market reflects this. The software product manager role has seen a 4x increase between 2014 and 2023 [1], which highlights the growing importance of software product managers in industry. With this increased significance, prescriptive literature describes that software product managers are often seen as *"the all-in-one job"* and generally, *"there is no single magic recipe for the definitions of responsibilities and the organization of software product management"* [33]. The ambiguous nature allows for the flexibility to maximize the product's value. However, organizational psychology research on team dynamics in the workplace shows that *role ambiguity*, defined as a lack of clarity in understanding the actions to be taken to achieve proposed individual goals [31],

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leads to poorly functioning teams [38]. Given this, we sought to uncover if there is misalignment in the expectations surrounding the software product manager role from the perspectives of the software product managers themselves and the software teams. As role ambiguity has a negative impact on *extra-role performance* [38], defined as actions not reflected in job descriptions that have an impact on organizational functioning [5], we reviewed job descriptions from industry and aligned the expectations therein against the role as described by industry practitioners. Role expectations identified by survey respondents that are absent or minimally represented from the job descriptions would suggest the presence of extra-role performance.

In addition to the potential negative effects of role ambiguity, effective communication skills are extremely important for software product managers due to their role as a liaison across the organization, software team, and customers [10–12, 15, 17, 33, 37, 57, 58]. After identifying the most frequent problems that software product managers face [59], researchers commonly proposed solutions [60] that include guidelines about transparent and consistent communication by translating the technical and non-technical aspects of the product to align the stakeholders. Without effective alignment and communication, contention between software product managers and stakeholders, especially software engineers, is inevitable.

To understand SPM role expectations and software team misalignment, our research contributions center around the following research questions:

RQ1 What <X> characterize(s) the involvement of a software product manager on a software team?
  (a) X = actions; abbreviated to $RQ1_{act}$ (i.e., direct lens of role expectations)
  (b) X = knowledge; $RQ1_{know}$ (i.e., indirect lens)
RQ2 How well do the actual job responsibilities of software product managers align with stated expectations?
RQ3 What problematic experiences do software teams report while working with software product managers?

To answer RQ1 and RQ3, we designed and ran a survey targeting individuals in the software industry — 45 product-focused and 35 engineering-focused respondents from 22 companies — about their perspectives on the software product manager role including role expectations ($RQ1_{act}$), knowledge expectations ($RQ1_{know}$), and problematic experiences (RQ3). To assess RQ2, we compared 37 job descriptions to our results from RQ1 to identify the presence of extra-role performance expectations (Table 4). From our results, we designed a comprehensive framework across the software development lifecycle to reduce misalignment, ambiguity, and contention for SPMs and the software team. The contributions of this work include:

(1) Eight overarching software product manager responsibilities (Table 2) to asses role expectations from the software team.
(2) A taxonomy of 81 questions that software teams typically ask (Table 3)
(3) Eight problematic experience categories when working with software product managers (Table 5)
(4) An evaluation of extra-role expectations comparing the software team and organizational perspectives (Table 4)
(5) The **S**PM **E**xpectations **N**ested in **S**oftware **E**ngineering (SENSE) framework which maps the role responsibilities, questions, and problems to the product development lifecycle (Section 7.3).

Software product manager (SPM) roles vary in titles across the industry [59]. For simplicity, in this paper, we use the acronym PF to refer to our product-focused respondents (e.g., software product manager, technical product manager, user experience). We use EF to refer to members within the software team (e.g., software engineer, engineering manager, data scientist, technical program manager).

2 Related Work

Software product management has evolved over the past twenty years [48]. This section describes the expectations surrounding the SPM role, and prior work concerning software team dynamics and information needs.

2.1 Software Product Management Expectations

Prior work describes the expectations of the SPM role from three main perspectives: the differentiation of the role, the impact of organizational factors on the role, and educational efforts to prepare students for this career path.

2.1.1 Role Expectations. Product management can be differentiated from other typical *PM* roles by thinking about expectations: *product managers* are analogous to *"results champions"* (i.e., maximizing business and customer impact) whereas *project managers* are comparable to *"implementation champions"* as they ensure the product is delivered on time [11]. Software, however, introduces another layer of complexity. A software product manager is a multidisciplinary role that engages many stakeholders, understands the market and users, all while prioritizing the necessary functionality [58]. The Software Product Management Body of Knowledge (SPMBok) [33], adapts the Product Management and Marketing Body of Knowledge [21] for companies where software is their primary product. The SPMBok provides insights into the difficulties of having a software product, typically, without any tangible pieces, while proposing a new framework that outlines the core responsibilities of an SPM across the functional areas within a typical software organization [33].

Boundary spanning, described in organizational psychology as integrating inter-group knowledge [43], is a critical part of the SPM role [36, 43], differentiating it from other *PM* roles, as they bridge diverse groups during software development [44]. With the inherent nature of software products, customers are often located across the globe, so an SPM's boundary spanning capabilities becomes an essential link between the customer's needs and the product's design [22]. However, without effective execution, this can lead to contention between the SPM and software team. This is another one of our study's contributions.

SPMs are found to increase quality of their project. In an analysis of project reports describing over 200 small to medium-sized projects, Ebert concludes that SPMs increase product quality (i.e., decreasing the number of defects found after release) and reduce schedule delays by over 80% [10]. Ultimately, the SPM maintained accountability across stakeholders to stabilize the projected schedule and commitments. Brinkkemper and Ebert later interviewed practitioners from large software companies to define four factors, surrounding the importance of transparency and accountability of all stakeholders, for facilitating successful software product management [12].

Springer and Miler interviewed experienced SPMs (i.e., SPMs with at least five years of experience) on what specific challenges prevent them from fulfilling their responsibilities [59]. This study uses interviews (n = 10) and an industry survey (n = 89) to identify the perceived frequency and severity of 27 selected problems. Their results highlight that determining a product's value, working with other teams in silos, inconsistent priorities from stakeholders, and technical debt are among the most frequent and severe problems SPMs face. Along with Wróbel, a follow-up study uses focus groups to investigate how SPMs can overcome some of these previously identified problems [60]. For each problem, a set of solutions are outlined to highlight the typical guidelines observed and by using a Likert scale, how comprehensive and effective the solution is. Many of their proposed solutions involve the collaboration with the software team, yet their work does not share insight into how applicable these solutions are according to the software team.

Artificial Intelligence (AI) is evolving the SPM role, even creating a new role called *AI Product Managers* [47]. Through a systematic literature review, Parikh concluded that there are many powerful applications of AI (e.g., idea generation, market research) [48], however there is still a distinct problem with models generating falsified information [14]. Within companies, SPMs are using AI models to better understand large data sets [45], and this intersection of AI and SPMs has been addressed through a small number of interviews [30] and case studies [45]. In 2025, one study surveyed and interviewed product managers on their current usages for AI at a large software company which include administrative tasks, development, and task delegation [67], but SPMs do not see AI as useful (at this time) for planning tasks [30]. These recent studies, along with ours, highlight the growing importance of understanding the product management landscape in the age of AI.

2.1.2 Organizational Factors. The context of an organization dictates many aspects of an SPM's role. Using grounded theory, researchers explored how the SPM's responsibilities differ across company sizes and business environments. Agile methodologies impact an SPM's role, [65] making it similar to that of a *Product Owner* [66]. Especially in smaller teams, these roles are often used interchangeably, or an SPM assumes the responsibilities of a product owner. However, Kittlaus concludes that this does not scale effectively [32]. Researchers have also sought to understand how startup culture affects the role using grounded theory [42], how roadmapping is done in small companies through case studies [68], the impact of situational factors on SPM processes [4], and developed a pragmatic framework for SPMs working specifically at startups [50].

To explore hiring expectations, Sipos and Szabó analyzed job descriptions across the Hungarian software industry. Their results conclude that an SPM's core responsibilities involve managing the *roadmap, product strategy, stakeholders, development life cycle* and *priorities*. They also concluded that competencies surrounding *leadership*, *communication*, and *empathy* are essential for successful SPMs. Similarly, a survey with nearly 1000 responses, conducted by a popular software product management podcast, concluded that *communication, execution,* and *product sense* are the most valuable skills, but *empathy* was ranked relatively low [52]. This survey also examines that *showing significant business impact* is one of the best ways to get promoted. Pertaining to product management across industries, prior work has used dynamic capabilities, defined as distinct processes surrounding product strategy and development [34], to clarify how competencies influence hiring according to a company's characteristics [17]. Through this analysis, they created a framework targeting educators to influence software product management training material.

2.1.3 Educational Contexts. Pawar, et al. designed a software engineering graduate program with a product management concentration [51]. Similarly, Estes, et al. used an industry survey to propose 21 competencies for an undergraduate computer science software product management course, describing that the curriculum should center around acquiring *business knowledge, technical knowledge,* and *soft skills* [15]. They also identified what components of an SPM are most valuable; from both the SPM and software team's perspectives, *planning*, understanding the *context* about stakeholders (i.e., business, customers, and developers), and effective *communication* are the most important. However, the expectations of the software team with respect to their interactions with SPMs was not investigated.

2.2 Team Dynamics & Information Needs

Our study closely aligns with work identifying the information needs of software developers and team dynamics between the SPM and the software team. There have been several studies on what questions developers ask their teammates to coordinate work [18] and software evolution tasks [55, 56]. Zimmermann and Begel sought to identify questions that the software team wants to ask data scientists [3], which was later replicated at a different company [25]. However, none of these studies have focused on the questions that the software team wants to ask SPMs; this is one of our study's main contributions.

Research on software team dynamics stresses the importance of positive relationships and effective communication within the team [46]. The perception of team dynamics can even drastically affect job satisfaction and retention [28]. Prior work has reviewed how dynamics between software teams and *project managers* [69] and *agile project* management [27] affects product quality and decision making [40]. Other domains of research have uncovered how *project* management affects decision making [40] and empowerment within Agile teams [64]. Research has also focused on advocacy. This includes how engineering managers can advocate for their software team, or people who identify their role as a technical communicators can better align stakeholders by advocating for the user [39]. However, product management, and software product management specifically, are different disciplines with different challenges, and therefore worthy of study as well.

3 Methodology

Our study involves asking industry practitioners to describe the SPM role ($RQ1_{act}$), list questions they would want SPMs to answer ($RQ1_{know}$), and describe past problematic experiences when working with SPMs (RQ3). By collecting questions that respondents want answered by SPMs, we indirectly identify components of the SPM (i.e., a proxy to the required knowledge needed for the role). To gain insight into an organizational perspective, we analyzed job descriptions and subsequently compared them to the stated expectations from the practitioners. To answer RQ1 and RQ3, we ran a survey targeting roles in the software team which is discussed in Section 3.1. For RQ2, described in Section 3.2, we compare our findings from the survey with 37 job descriptions. Our results outline the skills that SPMs should have (RQ1), companies' expectations (RQ2), and problems that software teams encounter (RQ3).

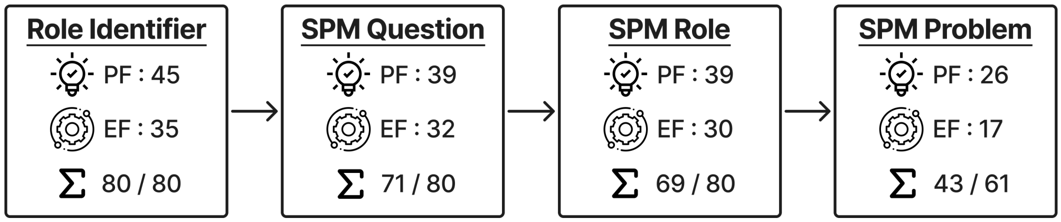
3.1 Survey

Our survey's design, approved by our institution's IRB (Protocol #26812), took inspiration from prior work that had similar goals: to understand how a role, in their case data scientists [3, 25], interact with software teams. We adapted the questions to our context of studying the SPM role and dug further into how the perceptions from product-focused (PF) and engineering-focused (EF) roles might differ.

3.1.1 Design. Since we want to compare the results from two populations, PF and EF, for Question 1, we filtered participants along one of three pathways based on their role: product-focused (e.g., software product manager, technical product manager, user experience), engineering-focused (e.g., software engineer, engineering manager, data scientist, technical program manager), or neither. If neither was selected, it was assumed that the participant's role is not relevant to our study and participants exited the survey. Figure 1 lists the survey questions and shows the flow for valid participants.

All roles were required to answer informed consent (before the survey) and Question 1. To distinguish the types of questions SPMs ask and answer, this resulted in two complementary questions: Questions 2a and 2b, which depend on the previously selected role identification question. Only the product-focused participants answered Question 2a and only the engineering-focused participants answered Question 2b. Questions 3, 4, and demography (described in Table 1) were answered by all participants.

3.1.2 Execution. The survey was created and managed through Qualtrics, and all data collected remained anonymous. Question 1 was the only required question. For recruitment, we posted on LinkedIn, alumni newsletters for our university, and contacted the authors' personal networks. To facilitate snowball sampling [49], where participants are



(1) **Role Identifier Question:** Within the software industry, are you in a more product-focused or engineering-focused role? [*single-response, required*]
   (a) Product-focused (e.g., software product manager, technical product manager, user experience)
   (b) Engineering-focused (e.g., software engineer, engineering manager, data scientist, technical program manager)
   (c) Neither / I'm not in the software industry [*exit*]
(2) **SPM Questions** [*up to 5 free-text, optional*]
   (a) (if *1a*) As a PM, what questions do you frequently answer for software teams?
   (b) (if *1b*) Suppose a software product manager (SPM) was added to your team. What requests would you make of the SPM and/or what information would you expect them to provide?
(3) **SPM Role:** In 1 - 3 sentences, how would you describe a software product manager's role within a team? *[1 free-text, optional]*
(4) **SPM Problems:** Describe a problematic experience you've had with a software product manager. [*1 free-text, optional*]

Fig. 1. Survey questions and flow. 'PF': *# product-focused participants*, 'EF': *# engineering-focused participants*, '$\sum$': *# total responses / # total shown* (e.g., 43 participants answered out of the 61 participants who saw Question 4 due to its later addition).

invited to join the recruitment process, a shortened URL was provided at the end of the survey so that participants could optionally forward it to others.

The researchers consulted with a software product manager and an industry researcher to refine the survey. At the end of the survey, an optional question allowed respondents to provide additional feedback during the pilot; this provided insight into potential pain points, leading to additional changes. The phrasing of the consent and questions, referenced in Figure 1, stayed the same throughout the evolution of the survey. In all, we ran the survey for 14 weeks and ended after a week of no responses.

Phase 1. The initial design did not include Question 4. There were 19 responses during Phase 1.

Phase 2. Based on feedback during Phase 1, we added seed questions to Question 2a and 2b. Similar to Begel and Zimmermann [3], we collected the seed questions from responses in Phase 1. The PF seed questions for Question 2a were:

- *"Should we go in this direction or that direction?"*
- *"What will customers think if we . . . ?"*
- *"Can you help me get clarity about . . . ?"*
- *"When should we prioritize . . . ?"*

Table 1. Optional Demographic Information. For all questions, the option *Prefer Not to Say* was included. Participants who selected this option were grouped with the respondents who did not respond. Percentages are calculated based on the total number of responses for each respective role and question. For example, the percentages in the *All* column for *What is the highest extent of your formal education?* are calculated out of 71, whereas the denominator in the *EF* column is 31.

Category	PF		EF		Σ All	
	#	%	#	%	#	%
How long have you been in your profession?						
< 1	1	3%	2	7%	3	4%
1 − 3	1	3%	4	13%	5	7%
3 − 5	3	8%	2	7%	5	7%
5 − 10	9	23%	5	17%	14	20%
10+	26	65%	17	57%	43	61%
Total	40	100%	30	100%	70	100%
What is the highest extent of your formal education?						
Some College	0	0%	3	10%	3	4%
Professional	1	3%	0	0%	1	1%
Associate's	0	0%	1	3%	1	1%
Bachelor's	16	40%	16	52%	32	45%
Master's	23	58%	6	19%	29	41%
Doctorate	0	0%	5	16%	5	7%
Total	40	100%	31	100%	71	100%
What is your gender identity?						
Female	10	26%	5	19%	15	23%
Male	28	74%	22	81%	50	77%
Non-binary	0	0%	0	0%	0	0%
Prefer to self-identify	0	0%	0	0%	0	0%
Total	38	100%	27	100%	65	100%

The EF seed questions for Question 2b were:

- *"What audience is this software for?"*
- *"What are the top user concerns with our system right now?"*
- *"What do customers need out of our software?"*

It also became clear through the responses that some respondents wanted to express explicitly what can go *wrong* with SPMs, and therefore Question 4 was added. In Phase 2, a Qualtrics collection issue caused us to lose one survey response, resulting in 21 applicable responses.

Phase 3. For Question 2a and Question 2b, we noticed participants often wrote only one question or skipped the final questions, likely due to survey fatigue. Thus, we reorganized the survey to require fewer clicks to navigate. There were 40 responses during Phase 3.

3.1.3 Participants. Our population consists of 80 software professionals — 45 of whom are product-focused and 35 are engineering-focused — who at least partially responded to the questions outlined in Figure 1. The demographic information was optional, and 71 participants of our population (89%) at least partially responded, as seen in Table 1.

Overall, many participants have five to ten years of experience (20%) and a majority have over ten years (61%). When looking at the distribution of education, 45% of the participants have a Bachelor's degree whereas 48% participants have some type of graduate degree (i.e., Master's or Doctorate).

In an optional question, participants reported to work for at least 22 companies: Amazon (including AWS), Anthropic, Bandwidth, Bank of America, Buildertrend, DaVinci Education Inc., Digital Turbine, DigitalOcean, Google, IBM, LexisNexis, Meta, Microsoft, Oracle, Pendo, Procter & Gamble, Prometheus Group, Redfin, Renaissance Computing Institute (RENCI), Rithum, SAS Institute, and Vistaly.

3.1.4 Analysis. To answer $RQ1_{act}$, $RQ1_{know}$, and RQ3, the research team followed a similar process used by the prior work [3, 25]. We chose reflexive thematic analysis (TA), outlined by Braun and Clarke [7], which is an entirely qualitative approach for verbal data analysis emphasizing that researchers interpret the *stories* within the data, rather than the identification of the *ground truth*. This analysis approach was chosen because we want to highlight that the responses are based on our respondents' lived experiences.

For this study, we completed three TAs: 1) Question 3 (*SPM Role*: $RQ1_{act}$), 2) Questions 2a and 2b (*SPM Questions*: $RQ1_{know}$), and 3) Question 4 (*SPM Problem*: RQ3). We merged the questions from PFs and EFs to ensure consistency across the TA. The research team agreed on an inductive analysis for each dataset so that latent codes emerge from the data [6, 63]. We also restricted final codes to appear at least three times throughout the dataset. Lastly, we report the number of unique participants per category, as we are interested in trends across the population, and this avoids putting too much weight on the responses of particularly verbose participants. Section 3.3 summarizes each TA including the number of responses, cards, discards, and results.

Data Cleaning and Preparation. To prepare the data, the first author read through all responses for relevance. For *SPM Questions*, a card represents *one* question. For *SPM Role* and *SPM Problem*, the first author split the compound responses to make the final set of *cards*. In this case, a card represents one main idea that a response discusses. For example, the response about the SPM role:

> *"They work with engineering and development teams to build features that meet market and customer needs. They also represent the software development team's interests to stakeholders and customers."*

was split into two separate cards:

a) *"They work with engineering and development teams to build features that meet market and customer needs."*
b) *"They also represent the software development team's interests to stakeholders and customers."*

Each card was labeled with a number and letter combination so that the number correlates to the response and the letter correlates to the number of parts. For example, the response above was the *5th* response and has two parts, so each part would be labeled as *5a* and *5b* respectively. The respondent type (i.e., PF or EF) was also included, but hidden from the research team during the card sorting process to limit any inherent bias. The final set of cards for each TA was then reviewed and approved by the second author.

Sorting the Cards & Thematic Analysis. This is an iterative and reflexive process [7]; for each TA, the first two authors collaboratively grouped similar cards. Each grouping was then given a name with a brief description to create our initial set of codes in our codebook. The codes were then refined, merged, and sometimes split to create our final set. After each change, the codebook was updated. Next, all authors met to converge on the final set of codes; this allowed the third author to come in without any preexisting notions to gain an outsider perspective on the categorization of the data. Lastly, the final codes for *SPM Role* and *SPM Problem* were grouped into overarching *themes*. Since the results from *SPM Questions* can indirectly identify responsibilities about the role, we will reference the *SPM Role* codes as *sub-themes* and the results from the *SPM Questions* TA as *codes* which fall under the *SPM Role* sub-themes (i.e., *SPM Question* codes ⇒ *SPM Role* sub-themes ⇒ *SPM Role* themes). A summary of each TA is described in Section 3.3.

Descriptive Question Identification. In prior work, after open card sorting, the researchers observed that many of their questions were either similar, too specific, or not concise which prompted the refinement of their responses into 145 descriptive questions (DQ) [3]. We observed something similar in the *SPM Questions* TA. For example,

- *"What is the strategic vision for this SW?"*
- *"How do we see this feature working in N years?"*
- *"What long term vision should we be aiming for?"*

were then combined into DQ66: : *What is the long-term vision for this product/feature?* To obtain a clearer view on unique questions within each code, we summarize and condense our initial 278 questions into 81 DQs. With this consolidated set of descriptive questions and their associated codes, our results offer more clear insights into what questions SPMs answer.

Differing perspectives of PF and EF was determined through the *population percentage difference* metric. Since the populations of PF and EF are independent and different sizes, converting the number of participants into a percentage of their population allows us to normalize the frequency for a DQ (similar to a test of two-proportions). For example, the DQ with the biggest misalignment, *"Why are we doing/building X?"* summarizes responses from 11 PFs and 1 EF. The calculated misalignment is found by $|\frac{11}{45} - \frac{1}{35}|$ which is 22%. The full list of descriptive questions can be found in Table 3.

3.2 Job Descriptions

When considering RQ2, the research team wanted to clarify software companies' stated expectations and compare them with the perspectives gathered from industry practitioners. Thus, we look for alignment between the collected survey data and publicly posted job descriptions. Ideally, we would compare against the stated internal job expectations, for example those used to assess for promotion. However, in the absence of access to those, job descriptions are a reasonable proxy. The research team followed a closed card sorting procedure (i.e., using a predefined codebook) [53] using the superset of *SPM Role* sub-themes and *SPM Questions* codes. The authors treated the *SPM Questions* codes as a granular perspective on the *SPM Role* sub-themes, adding a *"catch-all"* or default category for cases when there is not a clearly associated *SPM Question* codes. For example, if a card falls under the Understanding Customers sub-theme but does not explicitly mention one of the child codes – Who? or Problem – the card was labeled with the *"catch-all"* sub-theme: Understanding Customers. This maps the respondents' direct perception of *what* an SPM does to an employers' expectations. If extra-role performance is present, a sub-theme or code would not be present within the job descriptions analyzed; similarly, minimal representation shows responsibilities that must be accomplished by an SPM beyond the original stated expectations. To prepare the data, a researcher followed the same *preparation phase* described in Section 3.1.4. The research team adapted the typical closed card sorting procedure by allowing the two researchers who card sorted to create new codes. This was a necessary step for the research as any new codes suggest new topics not covered in the superset from RQ1.

3.2.1 SPM Jobs. To *verify* our codebooks, we collected job descriptions, primarily from the companies' website or LinkedIn, for 14 out of the 22 companies listed in Section 3.1.3 totaling 25 job descriptions. The remaining eight companies were not currently hiring for an SPM role. For this set of job descriptions, two authors iteratively collaborated to reach a consensus on the final coding scheme. To *generalize* our codebooks, we collected 12 additional job descriptions from 12 unique software companies not included in our *verify* set of job descriptions. The analysis of this set was done independently by two authors and an inter-rater reliability score was computed [41].

3.3 Data

The TAs identify themes, sub-themes, and codes to answer $RQ1_{act}$, $RQ1_{know}$, and RQ3. The closed card sort, using the superset from RQ1, answer RQ2. A summary of the data to answer each RQ is described next.

3.3.1 ***SPM Role ($RQ1_{act}$).*** For Question 3, we received 69 responses, which were split up to yield 148 cards. The themes – PRODUCT EXPERT, USER ADVOCATE, AND COMMUNICATION GURU – emerged from the eight sub-themes (labeled as *R#.*), which are outlined in Table 2 and described in Section 4.

3.3.2 ***SPM Questions ($RQ1_{know}$).*** Questions 2a and 2b yielded 278 questions written by PFs (n=169) and EFs (n=109). Since this survey question was asked to indirectly identify the role of the SPM, the 15 codes (labeled as *Q#.*) were mapped to the sub-themes identified in *SPM Role*. This mapping provides foundation for what the SPM should be capable of communicating to others to ensure team alignment. The mapping of *SPM Questions* is also outlined in Table 2 and further described in Section 4. The set of DQs (labeled as *DQ#*) can be found in Table 3.

3.3.3 ***SPM Jobs (RQ2).*** The closed card sort analysis of 37 job descriptions created 303 cards (verify: 191, generalize: 112). No additional codes were created, suggesting our codebook sufficiently covers the responsibilities outlined by software companies. An *almost perfect* agreement of .92 was computed for the *generalize* set of job descriptions [41]. The results of this card sort can be found in Table 4. A discussion on our findings is further described in Section 5.

3.3.4 ***SPM Problems (RQ3).*** Question 4 was introduced in Phase 2 of the survey (per Section 3.1.2), so we had fewer responses. The 43 responses were split into 58 cards, and 6 were discarded for relevance, yielding 52 cards for analysis. The eight problem codes (labeled as *P#.*) are described in Section 6 and outlined in Table 5. The discarded cards, while not answering the survey question deserve further discussion (Section 7.5).

4 RQ1: Software Team Expectations of the SPM Role

To answer RQ1, survey participants explained the role of the SPM directly (Question 3 in Figure 1, *SPM Role*) and also indirectly through the questions SPMs answer (Questions 2 in Figure 1, *SPM Questions*). The TA for $RQ1_{act}$ can be summarized with three high-level themes: PRODUCT EXPERT, USER ADVOCATE, AND COMMUNICATION GURU.

For SPM_{know}, the *SPM Questions* TA, yielded 15 types (i.e., codes) of questions that SPMs should be knowledgeable about. By mapping the codes from SPM_{know} to SPM_{act}, shown in Table 2, this provides a gateway into where the answer might come from. For example, to answer questions revolving around WHO? (Q5) and PROBLEM (Q6), the SPM should have sufficient knowledge about understanding the product's customers (R3).

In addition to clarifying the role, $RQ1_{know}$ aims to identify the typical questions that SPMs are asked by the software team from two different perspectives (i.e., PF and EF). After the *SPM Questions* TA, a set of 81 descriptive questions were identified (as described in Section 3.1.4) and are shown in Table 3. While the majority of the question codes included responses from both PF and EF participants, the consolidation into DQs often show a semantic difference. This misalignment, identified by the *population percentage difference metric* highlights a challenge SPMs often have to overcome: understanding *who* they are talking to, and *how* that changes the context. This analysis offers insight into prevalent communication differences between PF and EF individuals.

This section is organized by the *SPM Role* themes. Each *role* sub-theme (labeled with *R#*), situated under a theme, includes: 1) a description, and 2) the *SPM Question* code or codes mapped to it. Each *SPM Question* code (labeled with *Q#*) is accompanied by a description, and observed differences between PF and EF responses with relevant examples.

4.1 Communication Guru

This theme, representing nearly 50% of our responses (PF: 49%, EF: 47%), emphasizes the importance of SPMs effectively leading and aligning various stakeholders. By translating technical jargon into business opportunities, or vice-versa, the SPM can articulate the product's various goals.

4.1.1 ***R1. Among Stakeholders*** *[15 PFs, 38% | 14 EFs, 47%].* The SPM manages various stakeholders – leadership, software team, customers – and must effectively relay information between them through presentations, meetings, and documentation. This category and the following *question* categories describe typical boundary spanning characteristics as they must balance the wants and needs of all stakeholders present.

Q1. Company Alignment [8 PFs, 21% | 4 EFs, 13%]. Questions about the larger business goals was the second smallest category for EFs in this card sort. Nonetheless, some did want clarification concerning the business' top priorities and if any new potential plans could affect their job. When considering the questions that PFs listed, the most common was, DQ17: *How does this decision align with the company's objectives and strategy?* which might be more attributed to questions from other stakeholders; being able to answer this question and ultimately coordinate with all stakeholders is a core part of their role [33, 58].

Q2. Team Support [6 PFs 15% | 3 EFs, 9%]. With the embedded nature of the SPM role, alignment with the software team is crucial. The most frequent question within this category for PFs is, DQ24: *Can you track the answer to X (e.g., recurring meetings, reviewing, internal SW usage, cross-functional collaboration)?* This directly relates to one of their primary responsibilities: coordinating and managing stakeholders. Whether by connecting the team to someone else or settling disputes, this category describes a unique aspect of the SPM role compared to other *PM* roles. They must consistently be on the same page with the software team to either mitigate roadblocks, find answers, or point them in the right direction. The EFs in this case ask, DQ30: *What do you need from the software team?* which further emphasizes the reciprocal nature of supporting one another.

Q3. PM Process [3 PFs, 8% | 3 EFs, 9%]. Sometimes, the software team wants to learn more about what the SPM does and how they do it. For example, an EF asked, DQ49: *How do you balance stakeholders requests, development time required, and other business priorities to determine appropriate priorities?* On the other hand, one of each participant group clarified, DQ81: *Have we encountered X in the past?* which highlights the importance of the SPM being a product expert and in how they approach obstacles.

Table 2. RQ1: SPM Role & SPM Questions. 'PF': # PRODUCT-FOCUSED PARTICIPANTS, 'EF': # ENGINEERING-FOCUSED PARTICIPANTS, '∑': # TOTAL RESPONSES, 'DQ': # DESCRIPTIVE QUESTIONS | Percentages are column percentages calculated from the **TOTAL** rows.

Role Sub-Themes & Question Codes	**PF**		**EF**		**Σ All**		**DQ**	
	#	%	#	%	#	%	#	%
4.1 Communication Guru Theme								
R1. Among Stakeholders	15	38%	14	47%	29	42%	–	–
Q1. Company Alignment	8	21%	4	13%	12	17%	6	7%
Q2. Team Support	6	15%	3	9%	9	13%	3	4%
Q3. PM Process	3	8%	3	9%	6	8%	4	5%
R2. Context	5	13%	1	3%	6	9%	–	–
Q4. Why?	19	49%	3	9%	22	31%	5	6%
Role Sub-total	19	49%	14	47%	33	48%	–	–
Question Sub-total	26	67%	12	38%	34	53%	18	22%
4.2 User Advocate Theme								
R3. Understanding Customers	17	44%	13	43%	30	43%	–	–
Q5. Who?	15	38%	10	31%	25	35%	7	9%
Q6. Problem	7	18%	9	28%	16	23%	4	5%
R4. Solution Shaping	6	15%	3	10%	9	13%	–	–
Q7. Success Metrics	8	21%	8	25%	16	23%	5	6%
R5. Product Support Post-Release	4	10%	3	10%	7	10%	–	–
Q8. Product Rollout	7	18%	8	25%	15	21%	6	7%
Q9. Feedback	4	10%	6	19%	10	14%	4	5%
Role Sub-total	17	44%	15	50%	32	46%	–	–
Question Sub-total	16	41%	19	59%	25	49%	26	32%
4.3 Product Expert Theme								
R6. Product Strategy	16	41%	9	30%	25	36%	–	–
Q10. Requirements	15	38%	11	34%	26	37%	8	10%
Q11. Product Vision	10	26%	8	25%	18	25%	7	9%
R7. Planning	11	28%	11	37%	22	32%	–	–
Q12. Priority	25	64%	9	28%	34	48%	10	12%
Q13. Timeline	12	31%	4	13%	16	23%	5	6%
Q14. Roadmap	2	5%	5	16%	7	10%	3	4%
R8. Engineering Solutions	5	13%	3	10%	8	12%	–	–
Q15. Constraints	3	8%	5	16%	8	11%	4	5%
Role Sub-total	15	38%	15	50%	30	43%	–	–
Question Sub-total	30	77%	12	38%	26	59%	37	46%
SPM Role Total	39	100%	30	100%	69	100%	–	–
SPM Questions Total	39	100%	32	100%	71	100%	81	100%

Table 3. Descriptive Questions. 'PF': # PRODUCT-FOCUSED PARTICIPANTS, 'EF': # ENGINEERING-FOCUSED PARTICIPANTS, 'M%': |PF% – EF%| | Categories are represented multiple times when they had multiple descriptive questions with high misalignment.

ID	Descriptive Questions	Question Codes	PF	EF	M%
1	Why are we doing/building X?	Q4. Why?	11	1	22%
2	What should we work on next for this [cycle or release]?	Q12. Priority	9	1	17%
3	What are the top user concerns and complaints with our system right now?	Q9. Feedback	0	5	14%
4	How should this [requirement, feature] be designed, work, and integrated?	Q10. Requirements	9	3	11%
5	When does X (e.g., feature, task) need to be done by?	Q13. Timeline	5	0	11%
6	Should the team prioritize X or Y?	Q12. Priority	6	1	10%
7	Can you clarify what X is and what the source of truth is?	Q4. Why?	6	1	10%
8	Who is our target audience and where do they come from?	Q5. Who?	1	4	9%
9	What is X's priority, and when should we start working on it?	Q12. Priority	4	0	9%
10	We are experiencing some challenges, so can we have some more time?	Q13. Timeline	4	0	9%
11	What are the most significant and urgent requests from our users?	Q12. Priority	0	3	9%
12	What are the main goals for this product?	Q11. Product Vision	0	3	9%
13	What are the "must-have" compared to the "nice-to-have" requirements?	Q10. Requirements	0	3	9%
14	What is on the product roadmap?	Q14. Roadmap	0	3	9%
15	What does the user research show about this feature or idea?	Q8. Product Rollout	5	1	8%
16	Can you describe the personas for this product?	Q5. Who?	2	4	7%
17	How does this decision align with the company's objectives and strategy?	Q1. Company Alignment	3	0	7%
18	Can you help with priorities and trade-offs to stay on time and in budget?	Q12. Priority	3	0	7%
19	How many customers will be/are impacted by X?	Q5. Who?	3	0	7%
20	What do the customers want?	Q5. Who?	3	0	7%
21	What is the entire timeline for completing this product?	Q13. Timeline	1	3	6%
22	What is the first thing that we should do?	Q12. Priority	4	1	6%
23	What trade-offs are under consideration between scope, quality, and timeline?	Q4. Why?	4	1	6%
24	Can you track the answer to X (e.g., recurring meetings, reviewing, internal SW usage, cross-functional collaboration)?	Q2. Team Support	4	1	6%
25	What are the environmental, technological, and platform-specific constraints?	Q15. Constraints	0	2	6%
26	What are the key intra-business pain points, and how can we solve them in the simplest, most cost-effective way?	Q15. Constraints	0	2	6%
27	What are the usage estimates?	Q7. Success Metrics	0	2	6%
28	How will the product evolve?	Q14. Roadmap	0	2	6%
29	What is a part of the backlog?	Q12. Priority	0	2	6%
30	What do you need from the software team?	Q2. Team Support	0	2	6%
31	What is the problem that this product is trying to solve?	Q6. Problem	4	5	5%
32	What are the business rules we need to enforce?	Q15. Constraints	2	0	4%
33	What is the product's customer experience compared to what the customer expected?	Q9. Feedback	2	0	4%
34	What is the sequencing and the entire list of priorities?	Q14. Roadmap	2	0	4%
35	What is the financial feasibility of this product or specific feature?	Q1. Company Alignment	2	0	4%
36	Do you approve of the design of the new feature? If not, what's missing?	Q11. Product Vision	2	0	4%
37	Is there a Product Requirements Doc (PRD), and where can I find it?	Q10. Requirements	2	0	4%
38	How do you fit features into the overall development of the roadmap?	Q3. PM Process	2	0	4%
39	Can you help support managing the availability and collaboration between engineering teams?	Q2. Team Support	2	0	4%
40	What choice will have the most impact on growing our customer base?	Q11. Product Vision	2	0	4%
41	What are the customers' most significant pain points, and how many are affected?	Q6. Problem	2	3	4%
42	What is our product market fit and release plan?	Q8. Product Rollout	2	3	4%
43	What are the business, legal, and partner-related constraints that will impact the resources allocated for this project?	Q1. Company Alignment	1	2	3%
44	What is our risk tolerance for this change?	Q8. Product Rollout	1	2	3%

Continued on next page

ID	Descriptive Question	Question Codes	PF	EF	M%
45	Can you tell me more about the MVP (e.g., release plan, outlined features, and customer requests)?	Q11. Product Vision	4	2	3%
46	Is there any new planned intent that may impact my work?	Q1. Company Alignment	0	1	3%
47	From the users' perspective, what are the most important features to focus on?	Q12. Priority	0	1	3%
48	What features are users not aware of?	Q9. Feedback	0	1	3%
49	How do you balance stakeholders requests, development time required, and other business priorities to determine appropriate priorities?	Q3. PM Process	0	1	3%
50	Over time, are our priorities changing for the right reasons, or is this eroding our vision?	Q11. Product Vision	0	1	3%
51	Are you familiar with the product?	Q10. Requirements	0	1	3%
52	Does our target audience match the current users, and are they being helped? If not, what do we need to fix?	Q6. Problem	0	1	3%
53	How do you learn about our users' needs and desires?	Q3. PM Process	0	1	3%
54	Is this an internal or external-facing product?	Q5. Who?	0	1	3%
55	How should we handle stakeholder feedback or requests that conflict with current priorities?	Q12. Priority	1	0	2%
56	Can you demonstrate the unhappy paths?	Q6. Problem	1	0	2%
57	What is our go-to-market plan?	Q8. Product Rollout	1	0	2%
58	There's an easier way to do X - will it work for you and the customer?	Q7. Success Metrics	1	0	2%
59	What does the next 3-6 months look like?	Q13. Timeline	1	0	2%
60	Why do we need to meet this deadline?	Q4. Why?	1	0	2%
61	Why should we focus on this problem right now?	Q4 Why?	1	0	2%
62	What is the intended impact on the customers and business after release?	Q7 Success Metrics	3	3	2%
63	What data is collected for this feature or release's key success metrics (KPIs)?	Q7 Success Metrics	2	1	2%
64	How should we research and validate product decisions?	Q5 Who?	2	1	2%
65	What is our support strategy (e.g., maintenance, headcount, escalation, education) for this project?	Q8 Product Rollout	2	1	2%
66	What is the long-term vision for this product/feature?	Q11 Product Vision	2	1	2%
67	Which direction should we go to match your dreams about this product?	Q11 Product Vision	2	1	2%
68	What is the test plan and the associated test cases?	Q10 Requirements	2	1	2%
69	What is the target date for delivering this product?	Q13 Timeline	2	1	2%
70	How was the feature successful in meeting its goals?	Q9 Feedback	2	2	1%
71	How will we measure success to verify we are doing the right thing?	Q7 Success Metrics	2	2	1%
72	How should we decide to price this product?	Q8 Product Rollout	2	2	1%
73	What are the requirements for this product?	Q10 Requirements	2	2	1%
74	Where are the user stories and customer-specified use cases?	Q10 Requirements	3	2	1%
75	To ensure a successful product launch, what communication must happen with other cross-functional teams?	Q1 Company Alignment	1	1	1%
76	What are the leadership's current top priorities?	Q1 Company Alignment	1	1	1%
77	What technologies should we prioritize using, based on our team's expertise and the project's needs?	Q15 Constraints	1	1	1%
78	What features should be prioritized on the backlog for grooming?	Q12 Priority	1	1	1%
79	What is the long-term support plan for this product?	Q8 Product Rollout	1	1	1%
80	Can we break down our work or feature into smaller chunks?	Q10 Requirements	1	1	1%
81	Have we encountered X in the past?	Q3 PM Process	1	1	1%

4.1.2 ***R2. Context*** *[5 PFs, 13% | 1 EFs, 3%].* The SPM should convey the forest from the trees when it comes to the user perspective. Similarly, data-driven decision making, outlined in R4 Solution Shaping provides additional context in why something is happening. This is one of the least represented categories across the *SPM Role* TA even though it is the foundation for developing a product's strategy, planning, and metrics.

Q4. Why? [19 PFs, 49% | 3 EFs, 9%]. Questions in this category clarify and describe *why* a decision is made. This category includes the most listed question from PFs (i.e., DQ1: *Why are we doing/building X?*), and is the smallest category of questions for EFs. We suspect that while the EFs did not list these specific types of questions, they are often asked as a follow-up. Alternatively, this category may highlight one of the primary sources of contention between the two types of roles. It suggests that frequent discussions should occur around why choices were made. As the SPMBok highlights, one of the primary goals of an SPM is to lead through influence rather than authority [33]. To do so, when the SPM communicates the plan and outlook of the product, there should be supported justification on *why* this course of action is the best way to go. Similarly, DQ7: *Can you clarify what X is and what is the source of truth?* represents the only question listed by EFs, and it is tied for the third most listed question for PFs; this question stresses that communicating succinctly is extremely valuable [15].

4.2 User Advocate

SPMs should know the customer's needs, priorities, and use cases to deliver a valuable product. By thoroughly understanding the target audience, the SPM defines what success looks like, and how to market and deliver the right product. Their constant interaction with the customers allows real-time feedback to guide the team's next steps. This theme represents 44% of PFs and 50% of EFs.

4.2.1 ***R3. Understanding Customers*** *[17 PFs, 44% | 13 EFs, 43%].* The SPM must act as the voice of the customer by knowing the target audience thoroughly, including their problems, needs, and desires.

Q5. Who? [15 PFs, 38% | 10 EFs, 31%]. Overall, a similar percentage of participants from each role (about one-third) is represented in this category, but there is little agreement on the types of questions to better understand a product's target audience. When EFs ask questions in this category, they relate more with understanding the typical users (e.g., DQ16: *Can you describe the personas of this product?*, DQ8: *Who is our target audience and where do they come from?*). Instead, the PFs list questions such as DQ19: *How many customers will be/are impacted by X?* or DQ20: *What do the customers want?* which focuses more on gaining insight into what the customer thinks concerning a product's design and how they will be impacted.

Q6. Problem [7 PFs, 18% | 9 EFs, 28%]. The EFs and PFs largely agree on the types of questions revolving around the initial pain point that the product aims to solve. However, DQ31: *What is the problem that this product is trying to solve?* and DQ41: *What are the customers' most significant pain points, and how many are affected?* are among the most listed questions for EFs. Rather than being told what the product should do, the EFs want to know about the problems the users are facing to better understand what the best course of action is for creating a development strategy.

4.2.2 ***R4. Solution Shaping*** *[6 PFs, 15% | 3 EFs, 10%].* This role owns the product and drives decisions to align with business goals and market trends. With the SPM's in-depth knowledge about the stakeholders and product, solutions and relevant metrics are justified by data.

Q7. Success Metrics [8 PFs, 21% | 8 EFs, 25%]. When considering software product management, a concise and quantifiable way to show a product's impact is through metrics [33]. PFs solely listed the question, DQ27: *What are the usage estimates?* However, the questions listed by both roles support this interpretation as they ask DQ63:*What are the key success metrics (KPIs), and the subsequent data being collected for it, for this feature or release?* and DQ70:*"How was the feature successful in meeting its goals?"* Ultimately, both PFs and EFs want to identify whether they are or are not doing the right thing.

4.2.3 ***R5. Product Support Post-Release*** *[4 PFs, 10% | 3 EFs, 10%].*

PFs should be capable of determining relevant metrics to define success, promote adoption and develop proper go-to-market strategies. Similar to Solution Shaping, this category and the following question categories are represented more often in the companies' requirements rather than the software team where they mention the importance of proper marketing, sales, and training.

Q8. Product Rollout [7 PFs, 18% | 8 EFs, 25%]. This category includes questions about delivering the product to the customer, which includes topics like market research, risk tolerance, and supporting the product after release. Overall, this is a relatively balanced category between the PFs and EFs. Some of the most popular questions include,DQ15: *What does the user research show aout this feature or idea?*, DQ42: *What is our product market fit and release plan?* and DQ65: *What is our support strategy (e.g., maintenance, headcount, escalation, education) for this project?* Ultimately, these questions highlight the importance of supporting the customers and maintaining the product until its retirement [21].

Q9. Feedback [4 PFs, 10% | 6 EFs, 19%]. This set of questions shows some differences in how PFs and EFs define feedback. While this category is the second smallest in the card sort, two of the five descriptive questions are listed in Table 3; questions from EFs ask,DQ3: *What are the top user concerns and complaints with our system right now?*, emphasizing the importance of the continuous evolution of a product. Whereas the only question that PFs listed was,DQ33: *"What is the product's customer experience compared to what the customer expected?"* This SPM question seeks clarity on meeting product expectations.

4.3 Product Expert

Responses suggest that the SPM needs to own, understand, and prioritize feature development throughout the product's lifecycle and align the product with the software team's perspective.

4.3.1 ***R6. Product Strategy*** *[16 PFs, 41% | 9 EFs, 30%].*

This is about the big picture. An SPM should own, manage, and articulate the vision for the final product – from ideation to release – and how the team will get there.

Q10. Requirements [15 PFs, 38% | 11 EFs, 34%]. The most common question for both populations concerning a product's requirements is, DQ4: *How should this [requirement, feature] be designed, work, and be integrated?* This question is also tied for the second most listed question for PFs across the dataset (23%). Similarly, PFs often listed questions correlating to the expectations of their job; questions concerning *minimal-viable products* and *customer-specified use cases* are indicative of delivering a valuable product to the customers. The EFs asked questions that were relatively balanced across this category, however DQ13: *What are the "must have" compared to the "nice-to-have" requirements?* highlights their priorities and potential misalignment surrounding delivering a product that is necessary and technically feasible; this premise references a source of contention discussed in Section 6.1.1.

Q11. Product Vision [10 PFs, 26% | 8 EFs, 25%]. The product vision pinpoints what the ultimate goal and long-term plan for the product is. While there is no significant misalignment between the two roles, this category describes the EFs' initial need of wanting feedback from the customer; this relates to the 4.2.3 Feedback category of wanting feedback after shipping. The product vision questions also show instances of differing vocabulary. The PFs asked DQ66: *What is the long-term vision for this product/feature?* compared to the EFs' question of DQ12: *What are the main goals for this product?* While these two questions are ultimately about the end state of the product, there is a clear disconnect in how the question is framed; the SPM phrases it around the big picture, compared to the EFs want a more concrete representation of a set of goals.

4.3.2 ***R7. Planning*** *[11 PFs, 28% | 11 EFs, 37%].*

The SPM coordinates the product's lifecycle by prioritizing tasks and deadlines. Their plan should directly reference the goals outlined in the product strategy.

Q12. Priority [25 PFs, 64% | 8 EFs, 28%]. These questions highlight the biggest difference between the two disciplines, with PFs reporting more questions in this category. According to PFs, the most frequent questions from the software team focuses on deciding what matters most at a given point in time, which includes ranking items and making trade-offs. The two most prevalent questions, DQ2: *What should we work on next for this [cycle or release]?* and DQ6: *Should the team prioritize X or Y?* showcase that PFs most often answer questions about what is next. On the other hand, three EFs list, DQ11: *What are the most significant and urgent requests from our users?* In this case, the EFs want to understand the priorities from the user perspective.

Q13. Timeline [12 PFs, 31% | 4 EFs, 13%]. PFs report that EFs typically ask questions such as, DQ5: *When does X (e.g., feature, task) need to be done by?* or DQ10: *We are experiencing some challenges, so can we have some more time?* However, the EFs really want to ask the PFs about the entire timeline for the product. This question hints at potential misalignment when it comes to the feasibility of deadlines, further discussed in Section 6.1.1, and the software team wanting the big picture on when work is expected to occur (i.e, start dates and deadlines).

Q14. Roadmap [2 PFs, 5% | 5 EFs, 16%]. Creating and managing the product roadmap is essential to the SPM role [33, 58], but PFs did not think EFs commonly asked these types of questions. Nearly a quarter of the EFs want to ask, DQ28: *How will the product evolve?* Similarly, they ask DQ14: *What is on the product roadmap?* This emphasizes the need for better communication and coordination between the SPM and EFs.

4.3.3 ***R8. Engineering Solutions*** *[5 PFs, 13% | 3 EFs, 10%].*

SPMs uses the software team's expertise to identify solutions and further product improvements.

Q15. Constraints [3 PFs, 8% | 5 EFs, 16%]. This category, considering the product's constraints, involves questions surrounding the conditions and limitations on the resources available. We observed that the EFs err on the side of resource allocation and technical constraints that directly impact their work; they asked questions like, DQ26: *"What are the key intra-business pain points, and how can we solve them in the simplest, most cost-effective way?"* and DQ25: *"What are the environmental, technological, and platform-specific constraints?"* This category has the smallest representation of PFs, and their questions focus on the business impact and the prioritization of specific technologies. This may be due to differences in their vocabularies.

RQ1: The *role* of the software product manager is to be a communication guru with various stakeholders, user advocate, and (often technical) expert on the product. Answering *questions* is a central part of their daily activities, most commonly revolving around prioritization, requirements, who the customers are, and why something is happening.

5 RQ2 Results: Organization Expectations of the SPM Role

Approximately 90% of the software product management job descriptions analyzed include responsibilities related to the *SPM Role* themes about being a COMMUNICATION GURU, USER ADVOCATE, and PRODUCT EXPERT, suggesting broad thematic alignment between organizational role definitions and software team expectations. However, our analysis also reveals important mismatches concerning specific responsibilities, offering insight into potential areas of extra-role

Table 4. *RQ2: Job Descriptions.* The *Jobs* column represents the number of unique job descriptions per specific category. The *All* column represents the number of unique job descriptions across the *SPM Role* and its children *SPM Question* codes. Percentages are column percentages calculated from the Total row (i.e., out of 37).

Sub-Themes & Codes	Jobs #	Jobs %	All #	All %
R1 Among Stakeholders	26	70%	33	89%
Q1 Company Alignment	12	32%		
Q2 Team Support	12	32%		
Q3 PM Process	11	30%		
R2 Context	2	5%	4	11%
Q4 Why?	2	5%		
R3 Understanding Customers	6	16%	15	41%
Q5 Who?	5	14%		
Q6 Problem	5	14%		
R4 Solution Shaping	12	32%	19	51%
Q7 Success Metrics	10	27%		
R5 Product Support Post-Release	9	24%	18	49%
Q8 Product Rollout	14	38%		
Q9 Feedback	5	14%		
R6 Product Strategy	19	51%	28	76%
Q10 Requirements	11	30%		
Q11 Product Vision	7	19%		
R7 Planning	12	32%	26	70%
Q12 Priority	12	32%		
Q13 Timeline	0	0%		
Q14 Roadmap	15	41%		
R8 Engineering Solutions	21	57%	22	59%
Q15 Constraints	2	5%		
Total	**37**	**100%**	**37**	**100%**

performance arising from SPM role ambiguity. This section describes these notable differences when comparing RQ1 to RQ2, and is organized by the $RQ1_{act}$: *SPM Role* themes. The superset's representation in *SPM Jobs* is shown in Table 4. For example, 32% reference the general importance of R4 Solution Shaping, 27% of jobs directly mention Q7 Success Metrics, and when considering the superset of the role sub-theme and and question codes, 51% unique jobs mention either.

Communication Guru. For R1, Among Stakeholders, 89% of all analyzed job descriptions referenced working with other stakeholders. However, only about a third explicitly mentioned *how* (i.e., the question codes) the SPM should interact with stakeholders. Cards that fell into this sub-theme, rather than the question codes, typically revolved around translating information between stakeholders, leading through influence rather than direct authority, training members of the product team, and maintaining documentation. An SPM's responsibility to convey CONTEXT (R2) is one of the least represented categories across the *SPM Role* TA and the job descriptions, 11%, even though it is the foundation for developing a product's strategy, planning, and metrics. Containing the third largest question category across the *SPM Question* TA (Table 2), this describes a primary area of extra role performance in that the processes for gaining context is not explicitly required of the SPM, but will vastly improve the alignment of stakeholders.

User Advocate. The UNDERSTANDING CUSTOMERS sub-theme (R3), representing 41% of jobs, also highlights a boundary spanning aspect of the role. Although, SOLUTION SHAPING (R4) is not represented as highly in the TAs, 51% of jobs reference the importance of developing a solution through rapid prototyping and metrics based on the current environment.

Product Expert. R6 PRODUCT STRATEGY and its question codes is the second most represented sub-theme from the job analysis (76% of job descriptions) where companies often reference the importance of end-to-end lifecycle management. PLANNING (R7) also has high representation appearing in 70% of the job descriptions. However, it largely depends on whether the company asks explicitly about priorities or roadmap. Jobs that mention planning often reference the SPM as an orchestrator as they are responsible for ensuring the product is built correctly and efficiently. The TIMELINE category (Q13) was not found in our job description analysis even though it represents 25% of our participants' questions. This hints at extra-role performance in requiring the SPM to complete tasks more often attributed to the engineering manager or project manager. An SPM's responsibility of ENGINEERING SOLUTIONS (R8) was mentioned in 59% of the job descriptions emphasizing the importance of scalability and performance, however only two companies emphasize the importance of characterizing the product's CONSTRAINTS (Q15). This could describe an organization's assumption that the development of solutions and requirements will be directly based off the necessary context of the product's constraints.

RQ2 Summary: Job descriptions emphasize stakeholder coordination but rarely specify concrete practices for this boundary-spanning role. There is little reference to working explicitly with the software team.

6 RQ3 Results: Problematic Experiences with SPMs

Among the challenges of working with SPMs, versus the challenges SPMs face (a topic covered extensively in prior work [59] and briefly in Section 7.5), we identified eight problem areas (labeled with *P#*), shown in Table 5, and converged on two themes: COMPREHENSION and COOPERATION. As the primary goal was to understand and resolve the ambiguity surrounding SPMs, each problem category will reference our results from RQ1. The percentages shown throughout this section are based on the number of participants who responded to this question: 24 PFs and 15 EFs (shown in Figure 5).

6.1 Comprehension

The SPM's knowledge about the product, customers, market, and overall plan largely affects team dynamics. This theme represents responses from the majority of our respondents (PFs: 58%, EFs: 67%).

6.1.1 P1. Technical Feasibility [6 PFs, 25% | 4 EFs, 27%]. This category describes a disconnect between the product's end goal and the practicality of the solution. For example, an EF describes that *"A complete lack of technical understanding*

Table 5. ⚠ *SPM Problem.* 'PF': # PRODUCT-FOCUSED PARTICIPANTS, 'EF': # ENGINEERING-FOCUSED PARTICIPANTS, '∑': # TOTAL RESPONSES | Percentages are column percentages calculated from the **TOTAL** row.

Theme	Code	PF		EF		Σ All	
		#	%	#	%	#	%
6.1 Comprehension	⚠ P1. Technical Feasibility	6	25%	4	27%	10	26%
	⚠ P2. Understanding the Product	2	8%	5	33%	7	18%
	⚠ P3. Unstable Prioritization	3	13%	2	13%	5	13%
	⚠ P4. Understanding the Customer	1	4%	3	20%	4	10%
	⚠ P5. Understanding the Market	2	8%	1	7%	3	8%
	Comprehension Total	14	58%	10	67%	24	62%
6.2 Cooperation	⚠ P6 Team Player	5	21%	6	40%	11	28%
	⚠ P7. Ego	2	8%	3	20%	5	13%
	⚠ P8. Role Creep	4	17%	0	0%	4	10%
	Cooperation Total	11	46%	9	60%	20	51%
	Total	24	100%	15	100%	39	100%

and vision causes [SPMs] to always take requirements (no matter how technically infeasible) from the stakeholders without helping craft the "requirement" into something that fits the near and long term vision of the product." The SPM needs to consult the software team when developing the 4.3.1 REQUIREMENTS, 4.3.2 TIMELINE, and the ultimate 4.3.1 PRODUCT VISION. Without understanding what is possible, and in this case, understanding what questions to ask the software team, the proposed product will be impractical. This source of contention in conjunction with the instances described in the TEAM PLAYER problem category provide insight into the foundational challenges between the two roles which is discussed further in Section 7.1.1.

6.1.2 P2. Understanding the Product [2 PFs, 8% | 5 EFs, 33%]. Unlike TECHNICAL FEASIBILITY, this category highlights instances where the SPM does not understand the current and already implemented product. For example, 17a: *"There have been multiple cases where the product manager clearly doesn't know or understand the requirements. So they try to assign tasks and judge work while not having a good understanding of the product."* Other participants described that a lack of technical background causes a misunderstanding on *"what needs to happen to keep the lights on…"* In support of this, from the *SPM Questions* card sort, an EF wanted to gauge an SPMs knowledge by asking *"Are you familiar with the product?"* This question is not frequent; however, it does offer insight that some EFs want to clarify the SPM's technical understanding.

6.1.3 P3. Unstable Prioritization [3 PF, 13% | 2 EFs, 13%]. The participants describe that the *product strategy*, *priorities*, and *requirements* are constantly changing and there seems to be no right direction. A PF even admits that, at the beginning of their career, they *"…frequently changed the scope of a project mid-development based on new inputs from stakeholders…The lack of a clear and consistent roadmap made it difficult for the team to deliver a cohesive product on time, impacting both morale and the final output."* This highlights that ultimately, the SPM often does try to do their best, but by attempting to accommodate everyone with last minute changes, it can create a hectic environment. Similarly, Springer and Miler also identify this as a common challenge for the SPMs [59]; potential solutions include using data-driven decision making to create a strategy, and to communicate any uncertainty with all relevant stakeholder groups [60].

6.1.4 P4. Understanding the Customer [1 PFs, 4% | 3 EFs, 20%]. The SPM does not gather enough information about the customer or target audience creating *"a product with [a] one dimensional view point."* Oftentimes, contention becomes apparent when the SPM decides what the customer wants without actually asking them. This category aligns with one of the most frequent problems discovered by Springer and Miler: *"determining the true value of the product"* [59]. In their follow-up work, they suggest incorporating a feedback loop with current and potential customers [60].

6.1.5 P5. Understanding the Market [2 PFs, 8% | 1 EFs, 7%]. Responses stress that there is a lack of differentiation of the product from their competitors and other business relations, creating a misalignment in the target market. A PF describes that a problem they have encountered stemmed from the SPM *"[n]ot understanding the broader context and causal relationships of a business."* While this is the smallest category in this card sort, it identifies key challenges that correlate with multiple question categories, such as PRODUCT ROLLOUT, FEEDBACK, and COMPANY ALIGNMENT. These related questions suggest that without getting feedback about the product from the users and greater business, the software team is unable to deliver something better.

6.2 Cooperation

As the primary liaison among stakeholders, an SPM's role heavily revolves around collaboration. This theme highlights that the majority of our participants (PFs: 46%, EFs: 60%) experienced problems when SPMs did not effectively work with others.

6.2.1 P6. Team Player [5 PFs, 21% | 6 EFs, 40%]. These responses are rooted in an SPM's lack of communication and not fostering relationships on trust. This communication should involve periodic updates, aligning the team towards a common goal, and to ultimately show ownership of the product. When issues arise around being a TEAM PLAYER, as an EF described, they *"misrepresented what our team was working on to other partner teams, causing a lot of problems for us."* Similarly, an SPM described that *"they're not bringing [the software] team along on the journey, which leads to incorrect assumptions and poor decision-making."*

6.2.2 P7. Ego [2 PFs, 8% | 3 EFs, 20%]. As a PF describes, there are instances of SPMs that *"are too bulish [sic] and self-assured."* Similarly, tension increases when choices are made that only benefit them self, not listening to the software team's opinions, or ultimately not making any decisions.

6.2.3 P8. Role Creep [4 PFs, 17% | 0 EFs, 0%]. This type of problem was only mentioned by PFs hinting that an SPM is expected to cover other roles' responsibilities which makes them less attentive to their current goal. For example, an SPM says,

> *"…Often time product management is task [sic] with "everything". Often times taking on the role of product, program and project manager. This is common in smaller companies and at times it is hard to draw a clear line, however; it can be seen when the attitude shifts from short and long term visions to day-to-day executions of software tasks."*

Organizational psychology research on team dynamics in the workplace shows that *role ambiguity*, defined as a lack of clarity in understanding the actions to be taken to achieve proposed individual goals [31], leads to poorly functioning teams and negative affects [38]. Given this, three of the four participants mention the similarities and overlap between the *project* and *product* manager roles. To support our findings, the SPMBok describes that the role, more often than not, is a *"universal caretaker"* to supplement other teams across the organization [33]. While we do acknowledge that the role is extremely context dependent, when considering the product and company, this role ambiguity has a negative impact on *extra-role performance* [38], defined as actions not reflected in job descriptions that have an impact on organizational functioning [5]. Similarly, according to prior work, SPMs often assume the role of the *product owner* within Scrum [19, 65, 66, 70], but this is ultimately not scalable [32].

In addition to the varied responsibilities of an SPM, one participant described that their SPM was actually *"a skilled project manager but hired as a product manager and succumbed to Maslow's Hammer."* The result of ambiguous guidelines created the situation of hiring the wrong *PM*, and without suitable guidance, the project manager did not transition into the responsibilities necessary for *product* management. While not in abundance, these cards highlight a pervasive issue at the organizational level where there is a need to standardize and clarify the role across a company, and more broadly, the software industry.

RQ3 Summary: The most prevalent issues when interacting with an SPM revolve around their lack of knowledge surrounding the current product, customers, market, and capabilities of the software team which creates misunderstandings of product legitimacy and vision. The Role Creep category in our data reflects the influence of role ambiguity on extra-role performance.

7 Discussion

The SPM's differentiating quality from other *PM* roles is defining and delivering value throughout the lifecycle of the *software* product. In this section, we will first align the results from RQ1 and RQ2 with prior work (Section 7.1), and with RQ3 (Section 7.2). We present the SPM Expectations Nested in Software Engineering (SENSE) framework by exploring the involvement of the SPM throughout a product's lifecycle (Section 7.3) and discuss some peripheral findings from our exploration of RQ1 in terms of current trends in software product management (Section 7.4) and other problems that may emerge through the SPM role, or lack thereof (Section 7.5).

7.1 Alignment with Related Work

From the *SPM Role* card sort, three of the four most predominant characteristics – communication among stakeholders (R1), understanding customers (R3), and being a product expert through planning (R7) – align with the SPM archetype developed by Springer and Miler [58]. Additionally, through the analysis of the SPM Hungarian job market, they listed the top five competencies as *roadmap management* (Q14 Roadmap), *development life cycle monitoring* (cross-cutting RQ1), *product strategy development* (R6 Product Strategy), *stakeholder management* (R1 Among Stakeholders), and *development task prioritization* (Q12 Roadmap). Our results echo these findings. However, we reveal that user advocacy (Section 4.2) is also a critical part of the role. Additionally, our results complement the findings of a product management podcast survey [52], as these skills directly map to the three main themes (i.e., product expert, communication guru, and user advocate) that we define as the core responsibilities of the SPM role.

7.1.1 ***Discussion of Problems with SPMs.*** We observed that the contention between the SPM and the software team roots from issues with 6.1 Comprehension and 6.2 Cooperation, however these themes are interdependent. While there are more instances surrounding the (lack of) knowledge that SPMs have, another area of contention stems from how the SPM interacts with others. SPMs must be willing to collaborate with the various stakeholders rather than making assumptions about the product, customers, and market (P7 Ego). An SPM's role largely depends on their authority within the company [37], and more importantly, should imbue value through influence rather than a hierarchical leadership style [33].

In prior work, to identify how an SPM provides value to a team, Estes, et al. [15] surveyed industry practitioners. Their results suggest value is created through *planning*, *understanding the product context*, and *communication*. When SPMs do not provide value in the ways that are expected, this creates problematic experiences. To understand how our work relates to the prior work, we map these value categories to the *SPM problems*. For example, if an SPM isn't providing value in terms of PLANNING, this could lead to UNSTABLE PRIORITIZATION:

(1) Value: *Planning* [15] ⇒ Problem: Unstable Prioritization
(2) Value: *Understanding the product context* [15] ⇒ Problem: Understanding the {Product | Customer | Market }, Technical Feasibility
(3) Value: *Communication* [15] ⇒ Problem: Team Player | Ego

We notice that only one of our problem categories, *Role Creep*, is not reflected in the prior work on an SPM's value [15], but it is reflected in other literature stating that an SPM is a *"multi-purpose person"* that often fills any gaps that could delay the product's success [33]. Likely, this perspective on SPMs is not well understood, as doing tasks outside their job description are likely to lead to product success, but may not be an explicit expectation of the role. Building a deeper understanding of this phenomenon is left for future work (Section 8.4).

7.2 Problematic Experiences and Organization Misalignment

From the organizational perspective (i.e., job descriptions), SPMs are expected to align stakeholders through communication and emphasizing consistency and efficiency. Software companies place strong emphasis on generating effective product strategies and creating data-driven solutions. Notably, however, job descriptions provide comparatively little guidance on *how* this work should be accomplished in practice.

The least represented *SPM Role* categories among job descriptions (i.e., appearing in fewer than 50% of descriptions) align closely with categories identified in *SPM Problem*, reinforcing the idea that role ambiguity directly affects software teaming. For example, only 38% of companies explicitly included responsibilities related to directly interacting with customers to better understand their needs (R3), and 49% emphasized gathering feedback or knowledge about the target market (R5). In the absence of these responsibilities, our respondents described situations in which SPMs were forced to make assumptions about product design rather than validating decisions with users, often resulting in less valuable products.

Across both $RQ1_{act}$: *SPM Role* and RQ3: *SPM Problem*, respondents emphasized the importance of developing solutions in close collaboration with the software team to maintain trust and ensure a shared understanding about the product and technical feasibility. Yet only 32% of job descriptions included responsibilities related to leveraging engineering expertise (R8). Finally, the least represented category, R2. Context, which appeared in only one job description, highlights a key distinguishing factor of effective software product management. SPMs must clearly communicating the rationale behind decisions in terms that stakeholders can understand, align with, and ultimately support.

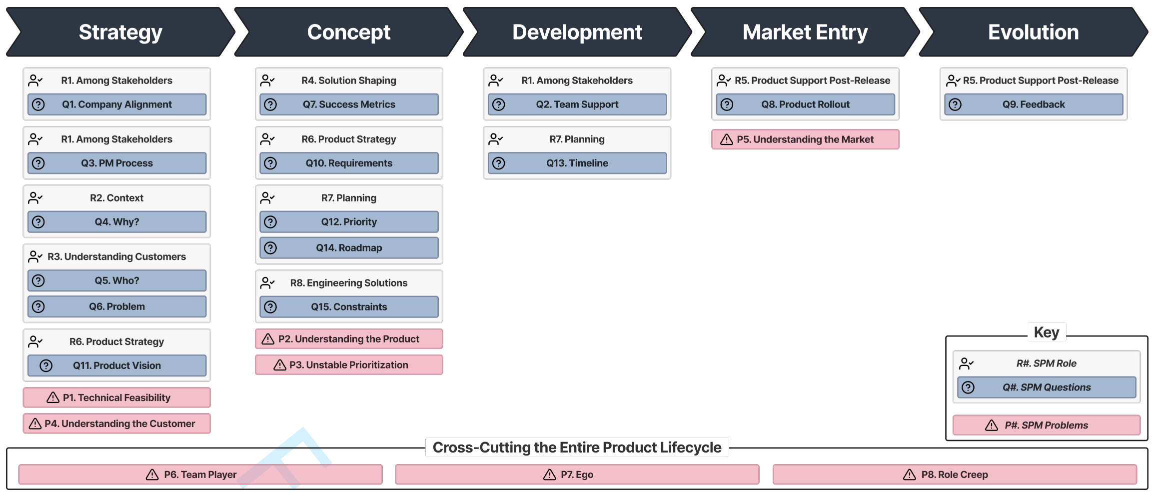


Fig. 2. The SENSE Framework. Categorization of the results from RQ1 (*SPM Role & Questions*) and the results from RQ3 (*SPM Problem*) into the five primary stages within the product lifecycle [12]. Each block represents the category's first appearance within the lifecycle but it can also occur in the following stages.

7.3 The SENSE Framework: SPM Expectations Nested in Software Engineering

Through the following discussion, we present the **S**PM **E**xpectations **N**ested in **S**oftware **E**ngineering (SENSE) framework for better alignment between the SPM and software team to support our goal of reducing role ambiguity and contention concerning the SPM. To do so, we clarify how the different role expectations and issues map to different stages of the software product lifecycle. As described by Ebert and Brinkkemper, there are five main stages of the typical product lifecycle: strategy, concept, development, market-entry, and evolution [12]. As a software product manager's role is to facilitate the development and production of a product from ideation to retirement, this lifecycle allows SPMs to typically gauge when to do something, the typical questions that will be asked, and who to interact with. Figure 2 maps our results from RQ1 and RQ3 onto the product lifecycle. Under each stage, codes from the *SPM Questions* TA with their respective *SPM Role* sub-theme appear. As the *SPM Job* description categories from RQ2 were sorted into the role categories, we do not model them explicitly. For example, under the *Strategy* stage, the SPM should be able to answer questions about the PROBLEM the product aims to solve (Q6) and WHO it is for (Q5). These questions are all about UNDERSTANDING CUSTOMERS (R3).

From RQ3, five out of the eight problems, all from the COMPREHENSION theme (Section 6.1), can also be attributed to specific stages of the product life cycle. Of these, the optimal way to reduce friction from these identified problems is by identifying what knowledge is missing; we recommend starting with the questions and aspects of the role listed under that same stage. For example, issues pertaining to UNSTABLE PRIORITIZATION (P3) are most related to the SPM's role of PLANNING (R7). The DQs outlined in Table 3 offer examples on the types of questions that can offer clarity.

While the majority of the *SPM Role* categories map to a particular stage, we identified that the three remaining *SPM Problem* categories, within the Cooperation theme (Section 6.2), cross-cut the product lifecycle, shown in the *Cross-cutting the Entire Product Lifecycle* portion of Figure 2. Two of these categories – P6 TEAM PLAYER and P7 EGO – identify systemic processes and issues that are not necessarily about the role but rather the person. Lastly, by more deliberately implementing product management processes or shared standards, the SPM can help mitigate the effects of role creep and ambiguity. On the whole, this framework serves as an intermediary between the SPM and software team to clarify expectations and identify missing knowledge, to ground discussions and align perspectives.

7.4 Current Trends in SPM

Both the Microsoft study identifying the most important questions for data scientists [3] and the subsequent replication at a financial technology company [25] had similar results showing that a data scientist requires similar knowledge across the software industry. When looking at RQ2, we found that while the job responsibilities are relatively stable across different companies, market conditions (and trends) can dictate the specifics of what is needed for new hires. Around 50% of the job descriptions analyzed for RQ1 indicate artificial intelligence, security, or cloud computing as additional requisite knowledge. Although SPMs may not be expected to have a detailed technical understanding of these trends, they must be cognizant of potential risks and opportunities associated with them [33].

As with trends in software engineering toward the use of AI to automate some engineering processes (e.g., [2, 13, 20, 54]), AI can also be useful for SPMs. Artificial intelligence can minimize the time spent finding various resources while maximizing the time spent analyzing that data, serving as a tool to speed up data-driven processes. Similarly, cloud computing helps identify an effective solution for a product that solves the user's problem with an efficient use of resources (Section 4.3.3). Moreover, with a large number of users and an increased potential for cyberattacks, security has become an important consideration for all products. As a result, when developing the product strategy (R6), specifically the requirements for the product, knowing about the inner workings of security allows the SPM to dedicate the right amount of resources to that component or iteration, especially during the testing and assessment of the risk tolerance of the product (R8). To support this, Weir et al. have used developer workshops to educate and engage SPMs in software security [71].

7.5 Discarded *SPM Problem* Cards

Prior to card sorting for *SPM Problems*, four cards were discarded on account of relevance (i.e., they did not describe a problematic experience with an SPM). However, they provide interesting perspectives.

One card, written by a EF, revealed that they did not work with SPMs, and explained that the lack of SPMs inappropriately placed the associated responsibilities onto engineering managers where their *"... valuable engineering skillsets [sic] aren't necessarily being used..."* because their time largely revolved around responsibilities often attributed to SPMs. This card did not reveal a specific problem they faced when working with an SPM, but rather highlighted the impact on the organization when the SPM ROLE CREEPS in the opposite direction, into the job responsibilities of the engineering manager. The other three cards were written by PFs and describe challenges they personally face in this role. These responses indicate that managing the expectations of stakeholders, leading through influence rather than authority, and not having enough engineers' support, all create challenges. These are reflective of Springer and Miler's prior work on the challenges of software product management [59].

8 Future Work

Based on our results, we identify several directions for future work.

8.1 Software Product Managers as Advocates

A successful SPM aligns stakeholders from the ideation to the retirement of the product, and for effective software product management, advocacy should be an inherent aspect of the role. To deliver a valuable product, they must advocate for the product's users. Our results (Section 4.2) and prior work [33, 39] have confirmed that user advocacy is important for team success. Similarly, in order to deliver a feasible product, they must advocate for the software team. Engineering managers who empower software teams increase team success [24], and we suspect the same for SPMs. Our results have identified that the most prominent problematic experiences with SPMs stem from this lack of advocacy for either the customers (i.e., P4 UNDERSTANDING THE CUSTOMERS) or the software team (i.e., P1 TECHNICAL FEASIBILITY, P6 TEAM PLAYER, P7 EGO). In addition to finding solutions to the problems faced by software teams, future work should identify how advocacy directly shapes the SPM role and their interactions with relevant stakeholders.

8.2 Within-Team Communication:

Our results from the survey and descriptive questions hint at some misalignment of priorities between SPMs and the software team. Additionally, although not fully comprehensive, we have identified eight sources of contention revolving around comprehension and cooperation (Section 6).

Our approach was a more general one, looking at trends across a variety of companies. We know from prior work that communication among developers is important to project success [23], but future research in that space should also consider different roles on the software team. Follow up work could include an analysis of within-team interactions between the software team and SPMs to identify specific points of miscommunication. Additionally, considering product and company characteristics within the analysis would offer insights into how the perceptions of the SPM role differ across multiple dimensions, such as customer characteristics (e.g., internal or external), product maturity (e.g., new to market, established), team size, and SPM seniority and experience.

8.3 Prioritization of Questions:

Through our exploration to answer RQ1, we generated a set of 81 descriptive questions that SPMs answer. In the study from which we derived our methodology [3], the researchers took the questions written by the software teams about what they would want to ask data scientists and asked engineers to rank those questions in terms of importance. We could do something similar and use a secondary survey to understand which questions members of the software team deem the most and least important. This analysis would lay the groundwork for common tasks and information that SPMs should know to best support the software team.

8.4 Clarification of Extra-Role Performance Expectations:

One problem identified solely by SPMs was about the ambiguous nature of the role and ROLE CREEP (Section 6.2.3). Organizational psychology research on team dynamics in the workplace shows that *role ambiguity*, defined as a lack of clarity in understanding the actions to be taken to achieve proposed individual goals [31], leads to poorly functioning teams and negative affect [38]. We would suggest interviewing SPMs to provide necessary context about the challenges and expectations they face in their day-to-day work, so those challenges can be translated into actionable strategies for education and practice.

8.5 Capitalizing on Opportunities for AI in Software Product Management:

Similar to how software engineers have started to incorporate these tools in their typical workflow [9, 29, 35, 62], we suspect that AI and LLMs have altered product management workflows, and case studies of five companies in prior work support this [45].

The results used to design the SENSE framework in this study were collected in 2023 when AI usage was still minimally used or integrated into workflows. Our participants were not prompted about AI usage. Especially when looking at the SPM role from an organizational psychology perspective, future work should further clarify the role and extra-role performance caused by the evolution of AI. Current usages for AI by SPMs include administrative tasks and development, but SPMs do not see AI as useful (at this time) for planning tasks [30]. Additionally, work on how SPMs delegate tasks at the individual, team, and organizational level within a large software company concludes that distinguishing between the responsibilities of the software team and the SPM will eventually overlap [67]. Future work should examine how the expectations from the software team outlined in our framework further evolves as AI usage increases.

Similarly, prior work has identified that SPMs currently distrust generative AI products [67]. Future work could also explore how to balance opportunities (e.g., productivity, aid in decision making, productivity) against the negative effects (e.g., legal, accountability, learning curve, trust) [30].

9 Threats to Validity

Conclusion Validity. Thematic analysis allowed the research team to identify relevant themes from the free-text questions listed in the survey. However, this process is collaborative and organic, offering multiple interpretations and conclusions. To mitigate this, multiple researchers were involved throughout this process, and to not skew our analysis, a card's corresponding role (i.e., PF or EF) was hidden until after the categories and themes were finalized. The late addition of Question 4 limited the breadth of answers. However, the themes and categories identified came from 16 companies and offer some preliminary insight into the collaboration issues between SPMs and the software team. For this paper, we do not consider the level of respondent and level of job description, so we cannot identify how extra-role expectations might differ at specific levels of the SPM role. This may explain why we did not see more of this phenomenon.

Internal Validity. Role names and responsibilities differ across companies, so the phrasing of our questions aimed to be generalizable across the software industry. Focusing on the general premise of the role (i.e., product-focused and engineering-focused), rather than specific titles may have been too narrow and led to misinterpretation. To provide more clarity, examples of some of the expected roles were included (e.g., software product manager, software engineer, engineering manager). We included seed questions, chosen directly from participants, for Question 2a and 2b. These questions were used in our analysis, and the categories that describe them correlate to higher popularity. Introducing these questions could have biased responses, thus limiting the breadth. However, this addition to our survey was based on the feedback from an industry researcher and experienced SPM.

External Validity. This survey aimed to maximize generalization across the software industry where we successfully targeted over 22 different software companies. However, this sample study offers limited opportunity to clarify answers resulting in lower realism [61]. To simulate a high level of realism, the inclusion of analyzing job descriptions from companies within and outside our respondents' pool of companies offers some verification into the generalizability and verifiability of our codebooks. The utilization of thematic analysis emphasizes the importance of using informative data rather than trying to reach saturation [8]. The diverse set of companies, and the comparison across datasets produces a rich picture of how to disambiguate the SPM role. We used snowball sampling [49] to recruit participants from our professional networks; we presume this skewed the participants' demography on the more experienced and technical side.

10 Conclusion

Software product managers are extremely valuable [10, 12] to software organizations for delivering valuable products to customers, and this research provides context to the expected and actual qualities necessary for SPMs to be successful. Software product management revolves around being a product expert who brings the product vision – created from understanding the customer – to fruition and provides the team with the context of planning and goals.

We identified 81 descriptive questions that software product managers typically answer. These questions accentuate that members of the software team want to understand the product's value beyond the code. While technically-focused questions were popular, revolving around the product's requirements and constraints, the engineering-focused roles deemed that information surrounding the users was equally as important; this includes wanting to know more about who the target audience is, *feedback* from the current users, and the *problem* the product is trying to solve.

While an SPM can be extremely valuable to a team, our study also highlights conflicts that they may have with the software team when they lack an understanding of the product, market, and customers. Other common negative experiences revolve around collaboration issues with the team, customers, or other stakeholders. Additionally, the team can also experience friction if the responsibilities required of the SPM extend beyond their role. As the theme, *collaboration* and its category, *role-creep* heavily revolve around communication, these are key indicators that the responsibilities of the software product manager role have ambiguous qualities. Our results, culminating in the SENSE framework, offer insight into the intricacies of SPMs working with software teams to ultimately disambiguate the knowledge and role expectations surrounding software product management.

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Table 6. RQ1: SPM Role & SPM Questions. 'PF': # PRODUCT-FOCUSED CARDS, 'EF': # ENGINEERING-FOCUSED CARDS, '$\sum$': # TOTAL CARDS | Percentages are column percentages calculated from the **TOTAL** rows.

Role Sub-Themes & Question Codes	PF		EF		Σ All	
	#	%	#	%	#	%
4.1 Communication Guru Theme						
R1. Among Stakeholders	22	25%	16	27%	38	26%
Q1. Company Alignment	8	5%	5	5%	13	5%
Q2. Team Support	7	4%	3	3%	10	4%
Q3. PM Process	3	2%	3	3%	6	2%
R2. Context	5	6%	1	2%	6	4%
Q4. Why?	24	14%	3	3%	27	10%
Role Sub-total	38	43%	18	31%	33	22%
Question Sub-total	42	25%	14	13%	40	14%
4.2 User Advocate Theme						
R3. Understanding Customers	19	21%	15	25%	34	23%
Q5. Who?	16	9%	11	10%	27	10%
Q6. Problem	7	4%	11	10%	18	6%
R4. Solution Shaping	7	8%	3	5%	10	7%
Q7. Success Metrics	8	5%	8	7%	16	6%
R5. Product Support Post-Release	4	4%	3	5%	7	5%
Q8. Product Rollout	12	7%	10	9%	22	8%
Q9. Feedback	4	2%	8	7%	12	4%
Role Sub-total	38	43%	18	31%	33	22%
Question Sub-total	27	16%	29	27%	27	10%
4.3 Product Expert Theme						
R6. Product Strategy	20	22%	11	19%	31	21%
Q10. Requirements	19	11%	14	13%	33	12%
Q11. Product Vision	13	8%	8	7%	21	8%
R7. Planning	14	16%	12	20%	26	18%
Q12. Priority	29	17%	11	10%	40	14%
Q13. Timeline	14	8%	4	4%	18	6%
Q14. Roadmap	2	1%	5	5%	7	3%
R8. Engineering Solutions	5	6%	3	5%	8	5%
Q15. Constraints	3	2%	5	5%	8	3%
Role Sub-total	38	43%	18	31%	33	22%
Question Sub-total	38	22%	18	17%	33	12%
SPM Role Total	89	100%	59	100%	148	100%
SPM Questions Total	169	100%	109	100%	278	100%

A Card Counts for SPM QUESTIONS, SPM ROLE, and SPM PROBLEM

The tables presented in the paper represent the unique participant count of each theme, sub-theme, and code to minimize unnecessary bias towards participants repeatedly talking about one subject. The tables in this section (Tables 6 & 7) display the card frequency in the three thematic analyses.

Table 7. Problem Codes. 'PF': # PRODUCT-FOCUSED CARDS, 'EF': # ENGINEERING-FOCUSED CARDS, '∑': # TOTAL CARDS | Percentages are column percentages calculated from the **TOTAL** row.

Theme	Code	PF		EF		Σ All	
		#	%	#	%	#	%
6.1 Comprehension	⚠ P1. Technical Feasibility	6	24%	5	19%	11	21%
	⚠ P2. Understanding the Product	2	8%	6	22%	8	15%
	⚠ P3. Unstable Prioritization	3	12%	2	7%	5	10%
	⚠ P4. Understanding the Customer	1	4%	4	15%	5	10%
	⚠ P5. Understanding the Market	2	8%	1	4%	3	6%
	Comprehension Total	14	56%	18	67%	32	62%
6.2 Cooperation	⚠ P6 Team Player	5	20%	6	22%	11	21%
	⚠ P7. Ego	2	8%	3	11%	5	10%
	⚠ P8. Role Creep	4	16%	0	0%	4	8%
	Cooperation Total	11	44%	9	33%	20	38%
	Total	25	100%	27	100%	52	100%

Data Availability

In our approved IRB protocol, participants only consented to release their responses in summary statistics and direct quotes provided in this paper. The full dataset of survey responses is unavailable. The survey, the job descriptions and associated job description card sort is available [16].

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